



Operating System

Lab 11

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Submitted to:

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TASKS

1. Write a Bash script that iterates through a list of planets and prints each planet's name along with a numerical count, starting from 1.

```
system@system-virtual-machine:~$ pico Task1
system@system-virtual-machine:~$ cat Task1
#!/bin/bash
count=1

for planet in Mercury Venus Earth Mars Jupiter Saturn Uranus Neptune
do
    echo "$count. $planet"
    ((count++))
done
system@system-virtual-machine:~$ chmod u+x Task1
system@system-virtual-machine:~$ ./Task1
1. Mercury
2. Venus
3. Earth
4. Mars
5. Jupiter
6. Saturn
7. Uranus
8. Neptune
```

2. Write shell script that print following output using while loop only.

```
system@system-virtual-machine:~$ pico Task2
system@system-virtual-machine:~$ cat Task2
#!/bin/bash

i=9

while [ $i -ge 0 ]
do
    j=$i
    while [ $j -ge 0 ]
    do
        echo -n "$j "
        ((j--))
    done
    echo # move to next line
    ((i--))
done
system@system-virtual-machine:~$ chmod u+x Task2
system@system-virtual-machine:~$ ./Task2
9 8 7 6 5 4 3 2 1 0
8 7 6 5 4 3 2 1 0
7 6 5 4 3 2 1 0
6 5 4 3 2 1 0
5 4 3 2 1 0
4 3 2 1 0
3 2 1 0
2 1 0
1 0
0
```

3. Produce above output again but this time using **for** loop.

```
system@system-virtual-machine:~$ pico Task3
system@system-virtual-machine:~$ cat Task3
#!/bin/bash

for ((i=9; i>=0; i--))
do
    for ((j=i; j>=0; j--))
    do
        echo -n "$j "
    done
    echo  # move to next line
done
system@system-virtual-machine:~$ chmod u+x Task3
system@system-virtual-machine:~$ ./Task3
9 8 7 6 5 4 3 2 1 0
8 7 6 5 4 3 2 1 0
7 6 5 4 3 2 1 0
6 5 4 3 2 1 0
5 4 3 2 1 0
4 3 2 1 0
3 2 1 0
2 1 0
1 0
0
```

4. Write a shell script that takes two numbers as argument from user like

./script 12 13

Write **add** function and pass these arguments to that function. Calculate sum and return result. Then display result at main script

```
system@system-virtual-machine:~$ pico Task4
system@system-virtual-machine:~$ cat Task4
#!/bin/bash

# Define a function to add two numbers
add() {
    local num1=$1
    local num2=$2
    local sum=$((num1 + num2))
    echo $sum    # Return the sum by echoing it
}

# Check if exactly two arguments are provided
if [ $# -ne 2 ]; then
    echo "Usage: $0 <num1> <num2>"
    exit 1
fi

# Pass arguments to add function and capture result
result=$(add "$1" "$2")

# Display the result
echo "The sum of $1 and $2 is: $result"
system@system-virtual-machine:~$ chmod u+x Task4
system@system-virtual-machine:~$ ./Task4 12 13
The sum of 12 and 13 is: 25
```





